



BAULKHAM HILLS HIGH SCHOOL

2014
YEAR 11 YEARLY

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions

Total marks – 70

This paper consists of TWO sections.

Section 1 – Multiple Choice 10 marks

Section 2 – Extended Response 60 marks

Attempt all questions

Start a new page for each question

Section 1 –Multiple Choice (10 marks)**Attempt all questions.****Answer the following on the booklet provided.****1** The number 0.09095 rounded to 2 significant figures is :

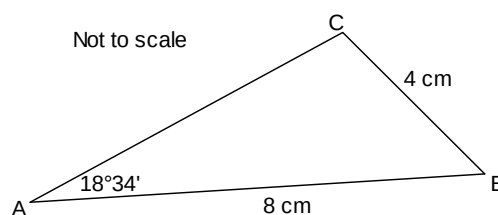
- (A) 0.10 (B) 0.09 (C) 0.091 (D) 0.0910

2 Simplify $5\sqrt{3} + \sqrt{20} - 2\sqrt{12} + \sqrt{45}$

- (A) $\sqrt{5} - \sqrt{3}$ (B) $\sqrt{5} + \sqrt{3}$ (C) $5\sqrt{5} + 9$ (D) $5\sqrt{5} + \sqrt{3}$

3 If α and β are the roots of $15x^2 + 75x - 3 = 0$, then $\alpha + \beta$ is :

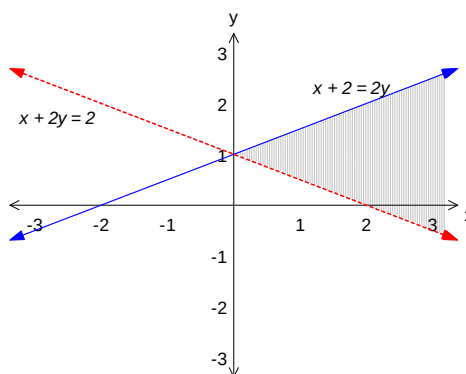
- (A) 75 (B) 5 (C) $-\frac{1}{5}$ (D) -5

4 A possible answer to the size of $\angle C$ in the triangle shown is :

- (A) $140^\circ 27'$ (B) $0^\circ 10'$ (C) $37^\circ 8'$ (D) none of these answers

5 What is the value of k if the expression $4x^2 - 6x + k$ is a perfect square ?

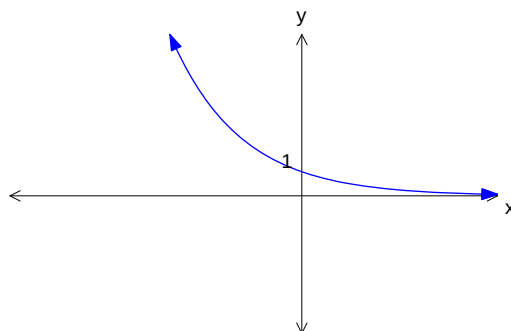
- (A) $\frac{4}{9}$ (B) $\frac{9}{4}$ (C) 4 (D) 9

6 The shaded region shown satisfies

- (A) $x + 2 \geq 2y$ and $x + 2y > 2$ (B) $x + 2 \geq 2y$ and $x + 2y < 2$
(C) $x + 2 \leq 2y$ and $x + 2y > 2$ (D) $x + 2 \leq 2y$ and $x + 2y < 2$

- 7 If $2 \cos x = \sqrt{3}$ in the domain $-180^\circ \leq x \leq 180^\circ$ then the values of x are :
- (A) $-60^\circ, 60^\circ$
 (B) $-120^\circ, 120^\circ$
 (C) $-30^\circ, 30^\circ$
 (D) $30^\circ, 150^\circ$

- 8 The graph illustrated could be :



- (A) $y = 2^x$
 (B) $y = (-2)^x$
 (C) $y = \left(\frac{1}{2}\right)^x$
 (D) $y = -2^{-x}$
- 9 The area of a rectangle with sides x and y is 47 cm^2 . The perimeter can be represented by:
- (A) $P = x + 47x^2$
 (B) $P = x + \frac{47}{x}$
 (C) $P = 2x + \frac{94}{x}$
 (D) $y = 2x + \frac{47}{x}$
- 10 Find the domain over which the curve $y = x^3 + 3x^2 - 24x + 7$ is concave downwards.
- (A) $x < -1$
 (B) $-4 < x < 2$
 (C) $x > -1$
 (D) $x < -4, x > 2$

End of Section I

Section II – Extended Response**All necessary working should be shown in every question.**

Question 11 (15 marks) - Start on the appropriate page in your answer booklet		Marks
a)	Solve $ 1 - 3x > 13$	2
b)	Find a and b such that $(\sqrt{3} + 4)^2 = a + b\sqrt{3}$	2
c)	For the function $y = \sqrt{9 - x^2}$	
	(i) What is the domain and range ?	2
	(ii) Is the function odd, even or neither? (Justify your answer by showing working)	2
d)	The directrix of a parabola is the x axis and the focus is the point $(1, 4)$. Find the equation of the parabola.	2
e)	Find the equation of the normal to the curve $y = 2x^2 - 5x + 1$ at the point where $x = 2$.	3
f)	Solve the pair of simultaneous equations $2x - y - 7 = 0$ $x + y + 1 = 0$	2
End of Question 11		

Question 12 (15 marks) - Start on the appropriate page in your answer booklet		Marks
a)	Simplify $\frac{2x^3-16y^3}{x^2+2xy+4y^2}$	2
b)	Differentiate (i) $\frac{2x+1}{x-1}$ (ii) $x\sqrt{x}$	2 2
c)	The function $y = f(x)$ is given by $f(x) = 3x(2x - 1)^2$ (i) Find the coordinates of the points where the curve $y = f(x)$ cuts the x axis. (ii) Find the coordinates of any stationary points on the curve $y = f(x)$ and determine their nature. (iii) Find any points of inflection. (iv) Sketch the curve $y = f(x)$ in the domain $-1 \leq x \leq 2$. (v) Hence find the maximum value for $y = f(x)$ in the given domain.	2 3 1 2 1
End of Question 12		

Question 13 (15 marks) - Start on the appropriate page in your answer booklet	Marks
a) The quadratic equation, $P(x)$, is given by $P(x) = x^2 - 2(k - 3)x + (k - 1)$ (i) Find the value(s) of k for which $P(x) = 0$ has distinct real roots. (ii) Explain what it means to the curve if $y = P(x)$ had been described as positive definite.	2 1
b) In the diagram $AB = 9\text{cm}$, $BC = 6\text{cm}$, $AD = 8\text{cm}$ and $AC = 12\text{cm}$. Not to scale (i) Find the size of $\angle ABC$, to the nearest degree. (ii) Find the area of $\triangle ABC$, to one decimal place. (iii) Given that $\angle ABC = \angle DAC$, prove that $\triangle ABC$ is similar to $\triangle CAD$. (iv) Hence find the ratio of the area of $\triangle ABC$ to the area of $\triangle DAC$	2 2 3 2
c) Prove that $\frac{\cos\theta}{1 - \tan\theta} + \frac{\sin\theta}{1 - \cot\theta} = \sin\theta + \cos\theta$	3
End of Question 13	

Question 14 (15 marks) - Start on the appropriate page in your answer booklet

a) Simplify

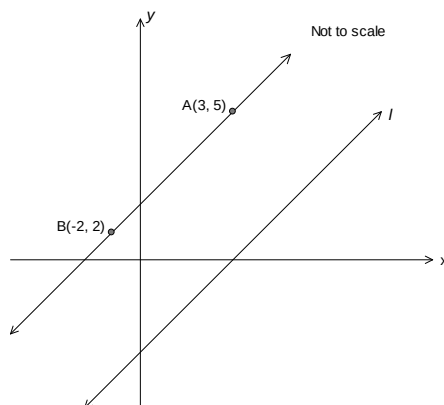
$$x^{-1}y^2(x^{\frac{1}{2}} - y^{-1})(x^{\frac{1}{2}} + y^{-1}) \text{ giving your answer with positive indices.}$$

2

b) Given $3x^2 + 4x + 5 \equiv A(x + 1)^2 + B(x + 1) + C$, find the value of the constants A, B and C .

2

c) The line through $A(3, 5)$ and $B(-2, 2)$ is parallel to the line l , $3x - 5y - 8 = 0$



(i) Show that the equation of AB is $3x - 5y + 16 = 0$

1

(ii) C is the point $(1, -1)$ on line l . Find the perpendicular distance of C from the line joining A and B .

2

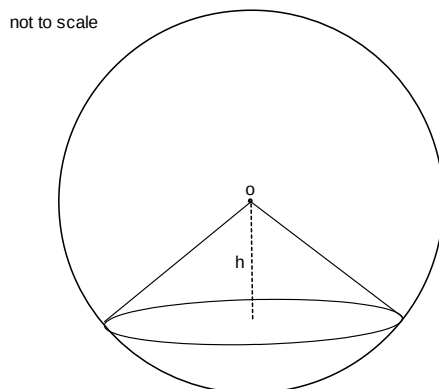
(iii) Find the area of the triangle formed by the points A, B and C .

2

(iv) Explain why the area of the triangle ABC is constant, regardless of the position of C on the line l .

1

d) A sphere contains a cone of height, h with its vertex at the centre of the sphere. The radius of the sphere is 12cm.



(i) Show that the volume of the cone is $V = \frac{\pi}{3}(144h - h^3)$

1

(ii) Given that the volume of the sphere is V_s , show that the maximum volume of the cone is $\frac{\sqrt{3}}{18} \times V_s$

4

End of Examination

Section 1. (10 marks)

1. C	4. A	7. C	10. A
2. D	5. B	8. C	
3. D	6. A	9. C	

10 marks

Section II

Quest 11. (15 marks)

a) $|1-3x| > 13$

$1-3x > 13 \quad 1-3x < -13$

$-3x > 12 \quad -3x < -14$

$x < -4 \quad \text{or} \quad x > \frac{14}{3}$

① mark each correct section.

2

b) $(\sqrt{3}+4)^2 = a+b\sqrt{3}$

$4+8\sqrt{3}+16$

$= 19+8\sqrt{3}$

$\therefore a=19 \quad b=8$

① mark each value

2

c) $y = \sqrt{9-x^2}$

i) $\mathbb{D}: -3 \leq x \leq 3$

$\mathbb{R}: 0 \leq y \leq 3$

ii) $f(x) = \sqrt{9-x^2}$

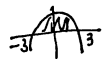
$f(-x) = \sqrt{9-(-x)^2}$

$= \sqrt{9-x^2}$

Since $f(x) = f(-x)$

 $f(x)$ is an even function.

$9-x^2 \geq 0$



① correct domain

① correct range from above working

① stating even

① working

4

d) $F(1,4) \quad a=2$
 $\therefore (x-h)^2 = 4a(y-k)$
 $(x-1)^2 = 8(y-2)$

① mark for finding 'a' or 'v'.

① for correct equation

2

e) $y = 2x^2 - 5x + 1$

$\frac{dy}{dx} = 4x - 5$

at $x=2 \quad m_1 = 8-5 = 3$
 $y = 2(2)^2 - 5(2) + 1 = -1$

$\therefore m_2 = -\frac{1}{m_1} = -\frac{1}{3}$
 $(2, -1)$

 $\therefore$ normal:

$y - y_1 = m(x - x_1)$

$y + 1 = -\frac{1}{3}(x - 2)$

$3y + 3 = -x + 2$

$x + 3y + 1 = 0$

① finding y value

① finding m_2

① forming equation

3

f) $\begin{cases} 2x - y - 7 = 0 \\ x + y + 1 = 0 \end{cases}$

$2x - y = 7 \quad \text{--- ①}$

$x + y = -1 \quad \text{--- ②}$

① + ②

$3x = 6$

$x = 2$

sub into ①

$4 - y = 7$

$-y = 3$

$y = -3$

$\therefore$ Soln is $x=2, y=-3$.

① correct x and y.

① off for each mistake

2

Question 12 (15 marks)

a) $\frac{2x^3 - 16y^3}{x^2 + 2xy + 4y^2} = \frac{2(x^3 - 8y^3)}{x^2 + 2xy + 4y^2}$
 $= \frac{2(x-2y)(x^2 + 2xy + 4y^2)}{x^2 + 2xy + 4y^2}$
 $= 2(x-2y)$

① correct factorising
① answer [2]

b) i) $y = \frac{2x+1}{x-1}$
 $\frac{dy}{dx} = \frac{(x-1)(2) - (2x+1)(1)}{(x-1)^2}$
 $= \frac{2x-2-2x-1}{(x-1)^2}$
 $= \frac{-3}{(x-1)^2}$

① applying quotient rule
① answer.

ii) $y = x\sqrt{x} = x^{\frac{3}{2}}$
 $\frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}}$ or $\frac{3\sqrt{x}}{2}$

① changing index power
applying product rule
① answer. [4]

c) $y = f(x) = 3x(2x-1)^2$
i) $f(x) = 0$ - x intercepts
 $3x(2x-1)^2 = 0$
 $x = 0, \frac{1}{2}$ [0,0] [1/2,0]

① for each answer. (2)

ii) Stat. pts.
 $f(x) = 3x(2x-1)^2$
 $f'(x) = 3(2x-1)^2 + 3x \times 4(2x-1)$
 $= 3(2x-1)(2x-1 + 4x)$
 $= 3(2x-1)(6x-1)$
 $f'(x) = 0$ for stat pts
 $\therefore (2x-1)(6x-1) = 0$
 $x = \frac{1}{6}, \frac{1}{2}$

① correct differentiation
① two correct points

c) ii) cont.

$f''(x) = 6(6x-1) + 3 \times 6(2x-1)$
 $= 36x - 6 + 36x - 18$
 $= 72x - 24$

$x = \frac{1}{6}$	$x = \frac{1}{2}$
$y = 3 \times \frac{1}{6} (2 \times \frac{1}{6} - 1)^2$ $= 0.22$	$y = 3 \times \frac{1}{2} (2 \times \frac{1}{2} - 1)^2$ $= 0$
$(\frac{1}{6}, 0.22)$	$(\frac{1}{2}, 0)$
test: $f''(\frac{1}{6}) = 72 \times \frac{1}{6} - 24$ $= -12 < 0$ max. $(\frac{1}{6}, 0.22)$	test: $f''(\frac{1}{2}) = 72 \times \frac{1}{2} - 24$ $= 12 > 0$ min $(\frac{1}{2}, 0)$

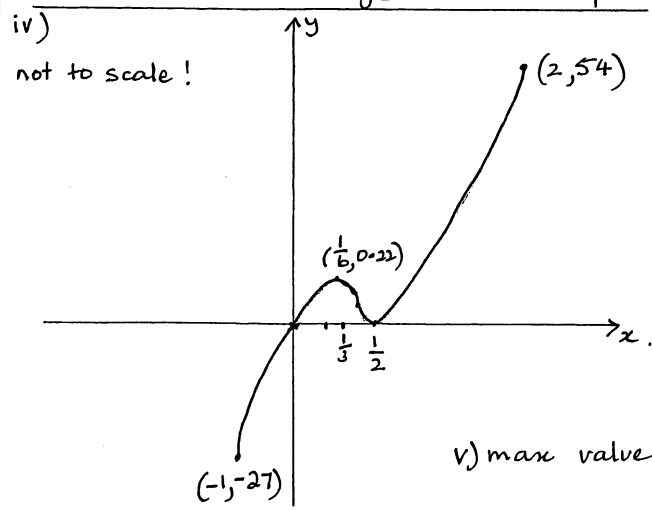
① for correctly testing both points

iii) P.O.I.

$f''(x) = 72x - 24 = 0$
 $x = \frac{1}{3}$
 $y = 3 \times \frac{1}{3} (2 \times \frac{1}{3} - 1)^2$
 $= 0.11$
 $(\frac{1}{3}, 0.11)$

① for correct POI and reason.

Test:
 $f''(\frac{1}{6}) = -12 < 0$ $f''(\frac{1}{2}) = 12 > 0$
 $\therefore$ a change in concavity.



① for end points
① clear diagram

v) max value $y = 54$

① for correct value. [9]

Question 13 (15 marks)

3) $P(x) = x^2 - 2(k-3)x + (k-1)$

i) $\Delta = b^2 - 4ac > 0$

$\therefore (-2(k-3))^2 - 4 \times (k-1) > 0$

$4k^2 - 24k + 36 - 4k + 4 > 0$

$4k^2 - 28k + 40 > 0$

$k^2 - 7k + 10 > 0$

$(k-5)(k-2) > 0$

$\therefore k < 2 \text{ or } k > 5$

ii) positive defn $\Delta < 0$

The curve $y = P(x)$ would be concave up and completely above the x axis

- ① forming quadratic
- ① correct solution

① logical explanation [3]

b) i) $\triangle ABC$

$\cos B = \frac{a^2 + c^2 - b^2}{2 \times a \times c}$

$= \frac{9^2 + 6^2 - 12^2}{2 \times 9 \times 6}$

$= \frac{-27}{108}$

B in 2nd quad.

$B = 180 - 75.52^\circ$

$= 104^\circ \text{ (nearest degree)}$

ii) $A_{\triangle ABC} = \frac{1}{2} ab \sin C$

$= \frac{1}{2} \times 9 \times 6 \times \sin 104^\circ$

$= 26.197 \dots$

$= 26.2 \text{ cm}^2$

- ① applying cosine rule correctly.

① correct angle

- ① applying area formula.

① correct answer

13. b) iii)

In $\triangle ABC$ and $\triangle CAD$

$\frac{BC}{AD} = \frac{6}{8} = \frac{3}{4}$ } same ratio

$\frac{BA}{AC} = \frac{9}{12} = \frac{3}{4}$

$\angle DAC = \angle ABC$ (given)

$\therefore \triangle ABC \parallel \triangle CAD$ (2 pairs of sides in ratio included $\angle$ equal)

① establishing ratios.

① angle.

① reason

iv) $A_{\triangle ABC} : A_{\triangle DAC}$

since $\frac{S_1}{S_2} = \frac{3}{4}$ (ratio found in proof)

then $\frac{A_1}{A_2} = \frac{S_1^2}{S_2^2}$
 $= \frac{9}{16}$

② using one of these methods to establish correct ratio.

or $A_{\triangle ABC} : A_{\triangle DAC} = \frac{1}{2} \times 9 \times 6 \times \sin 104 : \frac{1}{2} \times 8 \times 12 \times \sin 104$
 $= 9 \times 6 : 8 \times 12$
 $= 9 : 16$

or $A_{\triangle ABC} = 26.2$ (from above)

$A_{\triangle DAC} = \frac{1}{2} \times 8 \times 12 \times \sin 104$
 $= 46.57$

$\therefore 26.2 : 46.57 = 5625 : 100000$
 $= 9 : 16$

13 c)

= Aim: prove

$$\frac{\cos \theta}{1 - \tan \theta} + \frac{\sin \theta}{1 - \cot \theta} = \sin \theta + \cos \theta$$

$$\begin{aligned} \text{LHS} &= \frac{\cos \theta}{1 - \tan \theta} + \frac{\sin \theta}{1 - \cot \theta} \\ &= \frac{\cos \theta}{\cos \theta - \sin \theta} + \frac{\sin \theta}{\sin \theta - \cos \theta} \\ &= \frac{\cos \theta}{\cos \theta - \sin \theta} + \frac{\sin \theta}{\sin \theta - \cos \theta} \\ &= \frac{\cos \theta}{\cos \theta - \sin \theta} - \frac{\sin \theta}{\cos \theta - \sin \theta} \\ &= \frac{(\cos \theta + \sin \theta)(\cos \theta - \sin \theta)}{\cos \theta - \sin \theta} \\ &= \cos \theta + \sin \theta \\ &= \text{RHS} \end{aligned}$$

① off each error.

① linking LHS to RHS (concluding proof)

max 3 marks.

Question 14 (15 marks)

$$\begin{aligned} \text{a) } x^{-1} y^2 (x^{\frac{1}{2}} - y^{-1}) (x^{\frac{1}{2}} + y^{-1}) &= \frac{y^2}{x} (x - y^{-2}) \\ &= \frac{y^2}{x} \left(x - \frac{1}{y^2} \right) \\ &= \frac{y^2}{x} \left(\frac{xy^2 - 1}{y^2} \right) \\ &= \frac{xy^2 - 1}{x} \end{aligned}$$

① correctly expanding brackets

① correct final line.

2

$$\text{b) } 3x^2 + 4x + 5 \equiv A(x+1)^2 + B(x+1) + C$$

$$\text{RHS} = Ax^2 + 2Ax + A + Bx + B + C$$

$$= Ax^2 + x(2A+B) + A+B+C$$

$$\begin{aligned} \therefore 3 &= A & 4 &= 2A+B & 5 &= A+B+C \\ 4 &= 6+B & 5 &= 3-2+C \\ B &= -2 & 4 &= C \end{aligned}$$

$$\therefore A=3, B=-2, C=4$$

② for 3 correct values.

① for setting up equations and finding 1 correct value

2

c) AB.

(-2, 2)

$$1) m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$y - y_1 = m(x - x_1)$$

$$= \frac{5-2}{3+2}$$

$$y - 2 = \frac{3}{5}(x + 2)$$

$$= \frac{3}{5}$$

$$5y - 10 = 3x + 6$$

$$3x - 5y + 16 = 0$$

$$\text{ii) perp d} = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|3 \times 1 - 5 \times -1 + 16|}{\sqrt{9 + 25}}$$

$$= \frac{|3 + 5 + 16|}{\sqrt{34}}$$

$$= \frac{24}{\sqrt{34}} \text{ units}$$

① for establishing the equation.

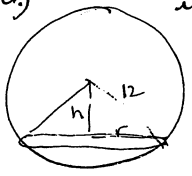
① correctly substitute into formula.

① correct answer.

$$\begin{aligned} \text{c. iii)} \quad \text{dist}_{AB} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{25 + 9} \\ &= \sqrt{34} \text{ units} \end{aligned}$$

$$\begin{aligned} A_{\Delta} &= \frac{1}{2}bh \\ &= \frac{1}{2} \times \sqrt{34} \times \frac{24}{\sqrt{34}} \\ &= 12 \text{ units}^2 \end{aligned}$$

iv) The line l , is a fixed distance from AB as it is parallel to AB . Therefore the perpendicular height is fixed and the base is fixed as A to B $\therefore$ Area is constant.

d)  i) $h^2 + r^2 = 12^2$
 $r^2 = 144 - h^2$
 $V = \frac{1}{3} \pi r^2 h$
 $= \frac{\pi}{3} (144 - h^2) h$
 $= \frac{\pi}{3} (144h - h^3)$

$$\begin{aligned} \text{ii)} \quad V_s &= \frac{4}{3} \pi r^3 \\ &= \frac{4}{3} \pi \times 12^3 \\ &= 2304 \pi \end{aligned}$$

$$V_{\text{cone}} = \frac{\pi}{3} (144h - h^3)$$

$$\frac{dV_c}{dh} = \frac{\pi}{3} (144 - 3h^2) \quad \frac{d^2V}{dh^2} = \frac{\pi}{3} (-6h)$$

$$\text{now } \frac{dV}{dh} = 0$$

$$\frac{\pi}{3} (144 - 3h^2) = 0$$

$$144 = 3h^2$$

① dist AB

① correct Area.

②

① the idea of parallel = fixed height

③

① for finding r^2

① correct derivative of V_c .

$$h^2 = 48$$

$$h = \pm \sqrt{48}$$

now $h > 0$

$$\begin{aligned} \therefore h &= \sqrt{48} \\ &= 4\sqrt{3} \text{ cm.} \end{aligned}$$

Test for max.

$$\begin{aligned} \frac{d^2V}{dh^2} &= \frac{\pi}{3} (-6h) \\ &= \frac{\pi}{3} (-6 \times 4\sqrt{3}) < 0 \\ \therefore \text{max.} \end{aligned}$$

$$\begin{aligned} \therefore V_c &= \frac{\pi}{3} (144h - h^3) \\ &= \frac{\pi}{3} (144 \times 4\sqrt{3} - (4\sqrt{3})^3) \\ &= \frac{\pi}{3} (576\sqrt{3} - 192\sqrt{3}) \\ &= \frac{\pi}{3} \times 384\sqrt{3} \\ &= 128\sqrt{3} \pi \end{aligned}$$

$$\therefore V_s = 2304 \pi$$

$$\begin{aligned} \frac{\sqrt{3}}{18} \times V_s &= \frac{2304 \pi \times \sqrt{3}}{18} \\ &= 128\sqrt{3} \pi = V_c. \end{aligned}$$

$$\begin{aligned} \text{or } \frac{V_c}{V_s} &= \frac{128\sqrt{3} \pi}{2304 \pi} \\ &= \frac{128\sqrt{3}}{2304} \\ &= \frac{\sqrt{3}}{18} \text{ as required.} \end{aligned}$$

① correct h.

① testing max/min

① finding concluding Volume ratio.

⑤